

Group17Turtlebot3proj.py – Changeable Parameter Reference

TurtleBot3 line-follower / barcode-navigation / obstacle-toppling controller. Every tuning constant lives in `__init__`. Parameters with a green environment-variable name can be overridden at launch without editing the file (e.g. `FORWARD_SPEED=0.12 python3 Group17Turtlebot3proj.py`); parameters marked — must be changed directly in the source.

Runtime environment variables (docstring-level)

Set before launch, e.g. `FORWARD_SPEED=0.12 python3 Group17Turtlebot3proj.py`. Any CAPITALS parameter built with `os.environ.get()` below can be overridden the same way.

Parameter	Default	Env. var.	What it controls
CONSOLE_STYLE	friendly	CONSOLE_STYLE	Console verbosity: 'friendly' (short, plain-English, one line/sec) or 'debug' (every number).
NO_COLOUR	0	NO_COLOUR	Set to 1 to disable ANSI colour in console output (e.g. when piping to a log file).
ENABLE_ACTUATOR	1	ENABLE_ACTUATOR	Set to 0 for a dry run: full routine runs but the actuator is never powered.
FAIL_STOP_HTML	1	FAIL_STOP_HTML	Set to 0 to stop writing the HTML fail-stop report.

GPIO pin assignments (BCM numbering)

Hard-coded pin numbers -- no environment override. Edit the source to change.

Parameter	Default	Env. var.	What it controls
LEFT_LED	23	—	GPIO pin driving the left indicator LED.
RIGHT_LED	24	—	GPIO pin driving the right indicator LED.
STOP_LED	25	—	GPIO pin driving the stop/red LED.
SERVO_PIN	18	—	GPIO pin driving the linear actuator (PWM, 50 Hz).

LEDs and buzzer

Parameter	Default	Env. var.	What it controls
BLINK_INTERVAL	0.30 s	—	Blink period for the ordinary lost-line alarm.
FAIL_STOP_BLINK_INTERVAL	0.07 s	FAIL_STOP_BLINK_INTERVAL	Very fast blink period used only for a latched fail-stop.
FAIL_STOP_REMINDER_SEC	15.0 s	FAIL_STOP_REMINDER_SEC	How often the fail-stop recovery reminder is reprinted.
WRITE_FAIL_STOP_HTML	True	FAIL_STOP_HTML	Whether the fail-stop card is also written as an HTML file.
FAIL_STOP_HTML_PATH	~/turtlebot3_fail-stop.html	FAIL_STOP_HTML_PATH	Path the HTML fail-stop report is written to.
TURN_LED_THRESHOLD	0.12	—	Line error magnitude above which the turn-direction LED lights.
BUZZER_INTERVAL	0.90 s	—	Minimum spacing between queued buzzer beeps.
BUZZER_SOUND_VALUE	1	—	TurtleBot3 Sound-service value used as the base buzzer tone.
FAIL_STOP_BUZZER_REPEAT_SEC	2.0 s	FAIL_STOP_BUZZER_REPEAT_SEC	How often the fail-stop siren pattern repeats.

Parameter	Default	Env. var.	What it controls
BEEP_BARCODE	{0:[1],1:[3],2:[4,4],3:[5,5]}	—	Beep pattern played for each decoded barcode value.
BEEP_RESCAN	[2, 2]	—	Beep pattern played when the obstacle routine re-scans.
BEEP_NAN_CLOSE_BOARD	[2,5,2,5]	—	Beep pattern for the close-board (too-near) NaN condition.
BEEP_FRONT_NAN_REVERSE	[5,2,5,2]	—	Beep pattern played while reversing off a too-close board.
BEEP_SIDE_CONFIRMED	[4]	—	Beep pattern once the obstacle side (left/right) is confirmed.
BEEP_ALIGNED	[5]	—	Beep pattern once fine alignment on the board is achieved.
BEEP_TACKLE	[5,4,5]	—	Beep pattern at the start of the push/tackle manoeuvre.
BEEP_SUCCESS	[4, 5]	—	Beep pattern for a successfully completed obstacle routine.
BEEP_FAILURE	[3,3,3]	—	Beep pattern for a recoverable failure.
BEEP_FAIL_STOP	[1,2,3,4,5,4,3,2,1,5,1,5]	—	Rising/falling siren reserved exclusively for a latched fail-stop.

Pixy2 image geometry

The Pixy2 line-tracking image is 79x52 'pixels'.

Parameter	Default	Env. var.	What it controls
FRAME_CENTER_X	39.0 px	—	Column where the line should sit; anything else is steering error.
TRACK_Y	45.0 px	—	Image row (near the bottom) at which the steering error is measured.
PIXY_LAMP_UPPER	1	PIXY_LAMP_UPPER	Pixy2 upper illumination LED on/off.
PIXY_LAMP_LOWER	1	PIXY_LAMP_LOWER	Pixy2 lower illumination LED on/off. Try 0 if glossy tape causes glare.

Normal line-following motion

Speeds are in metres/second unless noted.

Parameter	Default	Env. var.	What it controls
FORWARD_SPEED	0.16	FORWARD_SPEED	Normal line-following forward speed.
ARMED_FORWARD_SPEED	0.15	ARMED_FORWARD_SPEED	Forward speed once a turn command is armed, waiting for the junction.
CURVE_FORWARD_SPEED	0.14	CURVE_FORWARD_SPEED	Forward speed while taking a bend.
SLOW_FORWARD_SPEED	0.15	SLOW_FORWARD_SPEED	Reduced forward speed used in cautious phases.
CENTERING_SPEED	0.15	CENTERING_SPEED	Forward speed while re-centring on the line.
CURVE_FULL_SLOW_ERROR_PX	18.0 px	CURVE_FULL_SLOW_ERROR_PX	Line-error magnitude at which curve speed is fully reduced.
TURN_SPEED	0.95 rad/s	TURN_SPEED	Angular speed commanded during a junction turn.
MAX_TURN_SPEED	0.95 rad/s	MAX_TURN_SPEED	Angular speed cap during a junction turn.
kp (LINE_KP)	0.050	LINE_KP	Proportional gain of the line-following PD steering controller.
kd (LINE_KD)	0.012	LINE_KD	Derivative gain of the line-following PD steering controller.

Parameter	Default	Env. var.	What it controls
FORWARD_DISTANCE_AFTER_INTERSECTION_M	0.1 m	—	Distance driven forward into a junction before turning.
MOVE_CENTER_TIMEOUT_SEC	3.0 s	MOVE_CENTER_TIMEOUT_SEC	Timeout for driving into the junction centre.
TURN_TARGET_RAD	82 deg	—	Target turn angle for a junction turn (converted from degrees).
TURN_TOLERANCE_RAD	3 deg	—	Acceptable error on a completed junction turn.
MAX_TURN_TIME_SEC	4.0 s	—	Timeout for completing a junction turn.

Pixy vector filtering and intersection detection

Units are Pixy image pixels unless noted. No environment overrides.

Parameter	Default	Env. var.	What it controls
MIN_VECTOR_LENGTH_PX	10.0	—	Shortest line vector accepted as real (rejects noise specks).
MIN_FOLLOW_Y_SPAN_PX	7.0	—	Minimum vertical span for a vector to be used for line-following.
MIN_BOTTOM_Y_PX	18.0	—	Minimum bottom-Y for a vector to count as close to the robot.
MAX_ACCEPTED_ERROR_PX	38.0	—	Largest steering error accepted from a single frame.
MAX_LINE_X_JUMP_PX	26.0	—	Largest frame-to-frame jump in line X position accepted.
MIN_CROSSING_ANGLE_DEG	35.0	—	Minimum angle between two vectors to count as a crossing/junction.
CROSSING_SEGMENT_TOL_PX	5.0	—	Positional tolerance when matching crossing segments.
INTERSECTION_CENTER_X_TOL_PX	26.0	—	How far off-centre an intersection may be and still be accepted.
INTERSECTION_TRIGGER_Y_MIN	14.0	—	Lower Y bound of the intersection trigger zone.
INTERSECTION_TRIGGER_Y_MAX	51.5	—	Upper Y bound of the intersection trigger zone.
LINE_LOST_FRAMES	8 frames	—	Consecutive empty frames before the line is declared lost.
LINE_REACQUIRE_FRAMES	5 frames	—	Consecutive good frames needed to declare the line reacquired.

Barcode reading and voting

Controls how barcode reads are confirmed and how close together barcodes may be placed.

Parameter	Default	Env. var.	What it controls
BARCODE_VOTE_SAMPLES	3	BARCODE_VOTE_SAMPLES	Number of Pixy barcode reads collected per voting window.
BARCODE_MIN_VOTES	2	BARCODE_MIN_VOTES	Votes one code needs (of BARCODE_VOTE_SAMPLES) to be accepted.
BARCODE_VOTE_WINDOW_SEC	0.30 s	BARCODE_VOTE_WINDOW_SEC	Time window over which barcode votes are collected.
BARCODE_VOTE_SPEED	0.09	BARCODE_VOTE_SPEED	Forward speed while a barcode vote is being collected.
BARCODE_CLEAR_FRAMES	6 frames	—	Empty frames required before a barcode latch clears.

Parameter	Default	Env. var.	What it controls
INTERSECTION_CONFIRM_FRAMES	2 frames	—	Consecutive frames needed to confirm an intersection.
INTERSECTION_COOLDOWN_SEC	0.5 s	—	Minimum time between two accepted intersections.
BARCODE_ACCEPT_COOLDOWN	0.8 s	BARCODE_ACCEPT_COOLDOWN	Cooldown before the SAME barcode code may be accepted again.
BARCODE_ACCEPT_COOLDOWN_DIFFERENT	0.0 s	BARCODE_ACCEPT_COOLDOWN_DIFFERENT	Cooldown before a DIFFERENT barcode code may be accepted.
BARCODE_LATCH_CLEAR_DISTANCE_M	0.06 m	BARCODE_LATCH_CLEAR_DISTANCE_M	Distance travelled past a barcode before its same-code latch clears.
MAX_QUEUED_COMMANDS	2	MAX_QUEUED_COMMANDS	Maximum turn commands that may be queued awaiting their junction.
BARCODE_IGNORE_AFTER_COMMAND_SEC	0.00 s	BARCODE_IGNORE_AFTER_COMMAND_SEC	Blind period after any command is issued (0 = disabled).
POST_OBSTACLE_BARCODE_IGNORE_SEC	0.40 s	POST_OBSTACLE_BARCODE_IGNORE_SEC	Blind period for Barcode 2 only, right after the obstacle routine.
BARCODE2_REARM_CLEAR_FRAMES	12 frames	BARCODE2_REARM_CLEAR_FRAMES	Consecutive empty frames required before Barcode 2 may re-arm.
BARCODE2_REARM_DISTANCE_M	0.10 m	BARCODE2_REARM_DISTANCE_M	Distance from the saved obstacle start point before Barcode 2 may re-arm.
BARCODE2_ZONE_MIN_VOTES	3	BARCODE2_ZONE_MIN_VOTES	Votes needed for a non-2 code inside the Barcode-2 zone (unanimous).
DEBUG_VECTOR_LOG	False	—	Enable verbose per-vector debug logging.

Obstacle-routine geometry (measure on the physical robot)

These four numbers describe THIS robot's physical dimensions and should be measured with a ruler, not guessed.

Parameter	Default	Env. var.	What it controls
STOP_DISTANCE_M	0.20 m	STOP_DISTANCE_M	Distance from the board at which the robot stops before pushing.
LIDAR_TO_FRONT_EDGE_M	0.14 m	LIDAR_TO_FRONT_EDGE_M	LiDAR centre to the actuator tip, retracted (measured = 0.14 m).
ACTUATOR_REACH_M	0.05 m	ACTUATOR_REACH_M	Extra reach of the pusher tip when the actuator is fully extended.
TOPPLE_OVERTRAVEL_M	0.04 m	TOPPLE_OVERTRAVEL_M	Extra travel past first contact to push the board past its tipping point.
REAR_OVERHANG_M	0.14 m	REAR_OVERHANG_M	LiDAR centre to the back of the plate carrying the LED board.

Approach, tackle and return speeds

Parameter	Default	Env. var.	What it controls
APPROACH_SPEED	0.10 m/s	—	Base forward speed while approaching the board.
REVERSE_SPEED	-0.20 m/s	—	Reverse speed while returning to the saved start point.
RETURN_TOLERANCE_M	0.02 m	—	Position tolerance when returning to the saved start point.
REVERSE_SLOW_SPEED	-0.10 m/s	—	Reduced reverse speed near the return target.

Parameter	Default	Env. var.	What it controls
REVERSE_SLOW_ZONE_M	0.04 m	—	Distance from the target at which reverse speed is reduced.
REVERSE_STOP_MARGIN_M	0.04 m	—	Extra stopping margin during the return reverse.
REVERSE_LINE_CONFIRM_FRAMES	5 frames	—	Frames of line visibility needed to confirm reacquisition while reversing.
STOP_HOLD_DURATION	0.5 s	—	Time the robot holds stationary at the stop distance.
NUDGE_SPEED	0.06 m/s	—	Small forward speed used for close-range nudging.
BOARD_DOWN_THRESHOLD_M	0.35 m	—	LiDAR distance beyond which the board is considered toppled/gone.
ACTUATOR_EXTEND_HOLD_SEC	3.0 s	ACTUATOR_EXTEND_HOLD_SEC	How long the actuator is held extended.
TACKLE_PUSH_SPEED	0.04 m/s	TACKLE_PUSH_SPEED	Forward speed during the push/tackle manoeuvre.
TACKLE_MIN_PUSH_DISTANCE_M	0.04 m	TACKLE_MIN_PUSH_DISTANCE_M	Minimum push distance before the board can be judged toppled.
TACKLE_MAX_PUSH_DISTANCE_M	0.25 m	TACKLE_MAX_PUSH_DISTANCE_M	Maximum allowed push distance before aborting the push.
TACKLE_BOARD_DOWN_CONFIRM_SCANS	3 scans	TACKLE_BOARD_DOWN_CONFIRM_SCANS	Consecutive scans confirming the board is down.
TACKLE_CONTACT_HOLD_SEC	0.60 s	TACKLE_CONTACT_HOLD_SEC	Time to hold contact against the board once detected.
TACKLE_TIMEOUT_SEC	20.0 s	TACKLE_TIMEOUT_SEC	Overall timeout for the push/tackle state.
TACKLE_STALL_PROGRESS_M	0.004 m	TACKLE_STALL_PROGRESS_M	Odometry progress below which the robot is considered stalled (pushing).
TACKLE_STALL_CONFIRM_SCANS	4 scans	TACKLE_STALL_CONFIRM_SCANS	Consecutive scans confirming a stall/contact condition.
TACKLE_YAW_HOLD_KP	1.2	TACKLE_YAW_HOLD_KP	Proportional gain holding heading steady during the push.
TACKLE_YAW_HOLD_MAX	0.10 rad/s	TACKLE_YAW_HOLD_MAX	Angular speed cap while holding heading during the push.
RETURN_TIMEOUT_SEC	20.0 s	—	Overall timeout for the return-to-start manoeuvre.

Sensor freshness watchdog

Governs when a sensor interruption pauses motion versus triggers a latched fail-stop.

Parameter	Default	Env. var.	What it controls
SENSOR_STALE_TIMEOUT_SEC	0.75 s	SENSOR_STALE_TIMEOUT_SEC	Sensor data age at which it is considered stale and motion pauses.
SENSOR_RECOVERY_GRACE_SEC	3.0 s	SENSOR_RECOVERY_GRACE_SEC	Grace period allowed for sensors to recover before escalating.
SENSOR_RECOVERY_HEALTHY_CYCLES	3 cycles	SENSOR_RECOVERY_HEALTHY_CYCLES	Healthy watchdog cycles required before motion resumes.
SENSOR_MAX_RECOVERIES	2	SENSOR_MAX_RECOVERIES	Maximum recoveries allowed before a permanent fail-stop is latched.
FAIL_STOP_COMMAND_HOLD_SEC	1.0 s	FAIL_STOP_COMMAND_HOLD_SEC	Hold time applied to the last command around a fail-stop.

Parameter	Default	Env. var.	What it controls
WATCHDOG_PERIOD_SEC	0.05 s	WATCHDOG_PERIOD_SEC	Period of the independent sensor-health watchdog timer.
CONTROL_PERIOD_SEC	0.05 s	CONTROL_PERIOD_SEC	Period of the main Pixy control loop timer (20 Hz).

LiDAR range filtering

The LDS-02's specified minimum range is 0.16 m; readings below it are not treated as valid distances.

Parameter	Default	Env. var.	What it controls
SIDE_MIN_RANGE_M	0.16 m	SIDE_MIN_RANGE_M	Minimum valid side-sector LiDAR range.
SIDE_MAX_RANGE_M	0.60 m	SIDE_MAX_RANGE_M	Maximum valid side-sector LiDAR range.
FRONT_MIN_RANGE_M	0.16 m	FRONT_MIN_RANGE_M	Minimum valid front-sector LiDAR range.
FRONT_MAX_RANGE_M	0.60 m	FRONT_MAX_RANGE_M	Maximum valid front-sector LiDAR range.

Side-scan board-side voting

Determines whether the board is to the robot's left or right.

Parameter	Default	Env. var.	What it controls
SIDE_SCAN_COUNT	5 scans	—	Number of LiDAR scans taken for the left/right vote.
SIDE_REQUIRED_VOTES	3	—	Votes (of SIDE_SCAN_COUNT) required for a side to win.
SIDE_MAX_ATTEMPTS	2	—	Maximum rescan attempts if the vote is inconclusive.
SIDE_BOTH_DISTANCE_MARGIN_M	0.02 m	—	If both sides' distances differ by less than this, treat as tied and rescan.
SIDE_MAX_DISTANCE_SPREAD_M	0.30 m	—	Maximum acceptable spread among side-distance samples.
INITIAL_SCAN_SETTLE_DURATION_SEC	0.40 s	—	Settle time before the first obstacle-side scan.
RESCAN_SETTLE_DURATION_SEC	0.40 s	—	Settle time before a stationary rescan.

Close-board NaN classifier

Detects a board so close the LiDAR reports NaN instead of a distance.

Parameter	Default	Env. var.	What it controls
NAN_SIDE_MIN_RATIO	0.55	NAN_SIDE_MIN_RATIO	Minimum fraction of NaN returns on a side to count as 'board too close'.
NAN_SIDE_ADVANTAGE	0.30	NAN_SIDE_ADVANTAGE	Required NaN-ratio advantage of one side over the other.
NAN_VOTE_WINDOW	5	NAN_VOTE_WINDOW	Number of scans kept in the NaN-side voting window.
NAN_REQUIRED_VOTES	3	NAN_REQUIRED_VOTES	Votes required within the window to confirm a NaN-side result.

Front NaN handling (board too close after the turn)

Parameter	Default	Env. var.	What it controls
FRONT_NAN_HALF_ANGLE_DEG	25.0 deg	FRONT_NAN_HALF_ANGLE_DEG	Half-angle of the front sector examined for NaN returns.
FRONT_NAN_MIN_RATIO	0.50	FRONT_NAN_MIN_RATIO	Minimum NaN fraction in the front sector to trigger the reverse.
FRONT_NAN_CONFIRM_SCANS	3 scans	FRONT_NAN_CONFIRM_SCANS	Consecutive NaN-heavy scans required before backing away.
FRONT_NAN_EDGE_FALLBACK	True	FRONT_NAN_EDGE_FALLBACK	Treat an off-axis cluster as an edge artefact when the pusher axis itself is clear.
FRONT_NAN_REVERSE_DISTANCE_M	0.10 m	FRONT_NAN_REVERSE_DISTANCE_M	Distance reversed to clear a too-close board.
FRONT_NAN_REVERSE_SPEED	-0.05 m/s	FRONT_NAN_REVERSE_SPEED	Speed while reversing off a too-close board.
FRONT_NAN_RETURN_SPEED	0.05 m/s	FRONT_NAN_RETURN_SPEED	Speed while driving back forward after that reverse.
FRONT_NAN_HEADING_KP	1.5	FRONT_NAN_HEADING_KP	Heading-hold gain during the front-NaN reverse/return.
FRONT_NAN_MAX_ANGULAR_SPEED	0.15 rad/s	FRONT_NAN_MAX_ANGULAR_SPEED	Angular speed cap during the front-NaN reverse/return.
FRONT_NAN_REVERSE_TIMEOUT_SEC	5.0 s	FRONT_NAN_REVERSE_TIMEOUT_SEC	Timeout for the front-NaN reverse manoeuvre.

Rear geometry and rear-NaN handling

Parameter	Default	Env. var.	What it controls
REAR_NAN_REVERSE_MARGIN_M	0.03 m	REAR_NAN_REVERSE_MARGIN_M	Safety margin added to rear clearance before reversing.
REAR_NAN_INCREASE_BLOCK_RATIO	0.20	REAR_NAN_INCREASE_BLOCK_RATIO	NaN-ratio increase over baseline that blocks a reverse.
REAR_BASELINE_MATCH_TOLERANCE_M	0.03 m	REAR_BASELINE_MATCH_TOLERANCE_M	Tolerance for matching a rear reading to the robot's own self-view baseline.

Obstacle 90-degree turn

Stage 1 of the obstacle routine: an odometry-based turn to face the board.

Parameter	Default	Env. var.	What it controls
OBSTACLE_COARSE_TURN_DEG	90.0 deg	OBSTACLE_COARSE_TURN_DEG	Target turn angle to face the obstacle side.
OBSTACLE_TURN_TOLERANCE_RAD	3 deg	OBSTACLE_TURN_TOLERANCE_RAD	Acceptable error on the obstacle-facing turn.
OBSTACLE_TURN_SPEED	0.60 rad/s	OBSTACLE_TURN_SPEED	Angular speed for the bulk of the turn.
OBSTACLE_TURN_SLOW_SPEED	0.28 rad/s	OBSTACLE_TURN_SLOW_SPEED	Reduced angular speed as the turn nears its target.
OBSTACLE_TURN_CREEP_SPEED	0.10 rad/s	OBSTACLE_TURN_CREEP_SPEED	Slowest creep speed right at the end of the turn.

Parameter	Default	Env. var.	What it controls
OBSTACLE_TURN_SLOW_ZONE_RAD	22 deg	—	Angle-to-go at which the turn switches to slow speed.
OBSTACLE_TURN_CREEP_ZONE_RAD	8 deg	—	Angle-to-go at which the turn switches to creep speed.
OBSTACLE_TURN_TIMEOUT_SEC	6.0 s	OBSTACLE_TURN_TIMEOUT_SEC	Timeout for completing the obstacle-facing turn.

Front cluster detection and fine alignment

Stage 2: find the board cluster in the LiDAR scan and aim the pusher at it.

Parameter	Default	Env. var.	What it controls
FRONT_CLUSTER_HALF_ANGLE_DEG	45.0 deg	FRONT_CLUSTER_HALF_ANGLE_DEG	Half-angle of the sector searched for the board cluster.
FRONT_CLUSTER_MIN_POINTS	3	FRONT_CLUSTER_MIN_POINTS	Minimum LiDAR points required to count as a valid cluster.
FRONT_CLUSTER_MAX_ANGLE_GAP_DEG	3.0 deg	FRONT_CLUSTER_MAX_ANGLE_GAP_DEG	Maximum angular gap between points still counted as one cluster.
FRONT_CLUSTER_MAX_RANGE_JUMP_M	0.10 m	FRONT_CLUSTER_MAX_RANGE_JUMP_M	Maximum range jump between points still counted as one cluster.
FRONT_CLUSTER_MIN_WIDTH_M	0.015 m	FRONT_CLUSTER_MIN_WIDTH_M	Smallest cluster width accepted as a real board.
FRONT_CLUSTER_MAX_WIDTH_M	0.50 m	FRONT_CLUSTER_MAX_WIDTH_M	Largest cluster width accepted as a real board.
ACTUATOR_AIM_OFFSET_DEG	0.0 deg	ACTUATOR_AIM_OFFSET_DEG	Fixed bearing offset correction for actuator/LiDAR mounting misalignment.
FINE_ALIGN_SETTLE_DURATION_SEC	0.25 s	FINE_ALIGN_SETTLE_DURATION_SEC	Settle time before fine alignment scanning begins.
FINE_ALIGN_TOLERANCE_DEG	5.0 deg	FINE_ALIGN_TOLERANCE_DEG	Legacy centre-seeking bearing tolerance (ALIGN_MODE=centre).
FRONT_ALIGN_CONFIRM_SCANS	3 scans	FRONT_ALIGN_CONFIRM_SCANS	Consecutive agreeing scans required to confirm alignment.
FINE_ALIGN_KP	1.8	FINE_ALIGN_KP	Proportional gain of the fine-alignment rotation controller.
FINE_ALIGN_MIN_SPEED	0.045 rad/s	FINE_ALIGN_MIN_SPEED	Minimum rotation speed during fine alignment.
FINE_ALIGN_MAX_SPEED	0.18 rad/s	FINE_ALIGN_MAX_SPEED	Maximum rotation speed during fine alignment.
FINE_ALIGN_TIMEOUT_SEC	8.0 s	FINE_ALIGN_TIMEOUT_SEC	Timeout for the fine-alignment state.

Aim tolerance window (lateral-miss alignment)

The board is aimed at using lateral miss distance on the board face, not a fixed bearing angle. Set ALIGN_MODE=centre to restore the old strict centre-seeking behaviour.

Parameter	Default	Env. var.	What it controls
ALIGN_MODE	window	ALIGN_MODE	Alignment strategy: 'window' (lateral-miss, default) or 'centre' (legacy strict).

Parameter	Default	Env. var.	What it controls
BOARD_HIT_MARGIN_M	0.030 m	BOARD_HIT_MARGIN_M	Keep-off margin from each board edge inside the hit window.
BOARD_HIT_WINDOW_MIN_M	0.015 m	BOARD_HIT_WINDOW_MIN_M	Narrowest hit window ever accepted.
BOARD_HIT_WINDOW_MAX_M	0.060 m	BOARD_HIT_WINDOW_MAX_M	Widest hit window ever trusted, even for a wide board.
ALIGN_ENTER_FRACTION	0.60	ALIGN_ENTER_FRACTION	Fraction of the window required to ENTER alignment (hysteresis).
FINE_ALIGN_MAX_ANGLE_DEG	15.0 deg	FINE_ALIGN_MAX_ANGLE_DEG	Hard sanity cap on bearing error regardless of window size.
APPROACH_MAX_ANGLE_DEG	25.0 deg	APPROACH_MAX_ANGLE_DEG	Hard sanity cap on bearing error while approaching.
APPROACH_ROTATE_ONLY_FACTOR	2.0	APPROACH_ROTATE_ONLY_FACTOR	Miss size (in window-widths) at which the robot stops translating and rotates only.
APPROACH_AIM_MAX_SCANS	12 scans	APPROACH_AIM_MAX_SCANS	Scans at the braking distance before accepting any on-board aim as a last resort.
FRONT_SAFETY_STOP_M	0.17 m	FRONT_SAFETY_STOP_M	Absolute closest distance allowed before the robot must stop.

Approach to the board

Parameter	Default	Env. var.	What it controls
APPROACH_NEAR_DISTANCE_M	0.40 m	APPROACH_NEAR_DISTANCE_M	Distance at which the approach switches to near-board speed.
APPROACH_NEAR_SPEED	0.08 m/s	APPROACH_NEAR_SPEED	Forward speed once inside APPROACH_NEAR_DISTANCE_M.
APPROACH_AIM_TOLERANCE_DEG	4.0 deg	APPROACH_AIM_TOLERANCE_DEG	Legacy fixed-angle aim tolerance (ALIGN_MODE=centre only).
APPROACH_ROTATE_ONLY_DEG	9.0 deg	APPROACH_ROTATE_ONLY_DEG	Legacy rotate-only bearing threshold (ALIGN_MODE=centre only).
APPROACH_BALANCE_KP	1.8	APPROACH_BALANCE_KP	Proportional gain correcting bearing while approaching.
APPROACH_MAX_ANGULAR_SPEED	0.20 rad/s	APPROACH_MAX_ANGULAR_SPEED	Angular speed cap while approaching.
APPROACH_TIMEOUT_SEC	10.0 s	APPROACH_TIMEOUT_SEC	Timeout for the approach state.
APPROACH_MAX_DISTANCE_M	0.75 m	APPROACH_MAX_DISTANCE_M	Maximum plausible board distance; anything further is rejected.

Post-obstacle line reacquisition

Parameter	Default	Env. var.	What it controls
OBSTACLE_REACQUIRE_CONFIRM_FRAMES	2 frames	OBSTACLE_REACQUIRE_CONFIRM_FRAMES	Frames of confirmed line needed to leave the reacquire state.
OBSTACLE_REACQUIRE_SPEED	0.13 m/s	OBSTACLE_REACQUIRE_SPEED	Forward speed while reacquiring the line.
OBSTACLE_REACQUIRE_LARGE_ERROR_SPEED	0.10 m/s	OBSTACLE_REACQUIRE_LARGE_ERROR_SPEED	Reduced forward speed when the line error is large during reacquisition.

Parameter	Default	Env. var.	What it controls
OBSTACLE_REACQUIRE_MAX_ANGULAR_SPEED	0.70 rad/s	<code>OBSTACLE_REACQUIRE_MAX_ANGULAR_SPEED</code>	Angular speed cap while reacquiring the line.
OBSTACLE_REACQUIRE_SEARCH_SPEED	0.35 rad/s	<code>OBSTACLE_REACQUIRE_SEARCH_SPEED</code>	Angular speed while actively searching (line not yet visible).

Compatibility / legacy fallback

Parameter	Default	Env. var.	What it controls
CENTERING_BALANCE_TOLERANCE	0.060	—	Legacy left/right balance tolerance for the old fallback centring branch.
CENTERING_EXTRA_DURATION_SEC	0.0 s	—	Legacy extra centring duration for the old fallback branch.

Actuator pulse widths

Parameter	Default	Env. var.	What it controls
ACTUATOR_EXTEND_PULSE_MS	2.0 ms	<code>ACTUATOR_EXTEND_PULSE_MS</code>	RC-style PWM pulse width commanding the actuator to extend.
ACTUATOR_RETRACT_PULSE_MS	1.0 ms	<code>ACTUATOR_RETRACT_PULSE_MS</code>	RC-style PWM pulse width commanding the actuator to retract.

Console status logging and LiDAR diagnostics

Parameter	Default	Env. var.	What it controls
STATUS_LOG_INTERVAL	1.00 s	<code>STATUS_LOG_INTERVAL</code>	Minimum interval between printed status lines.
BLIND_VALUE_MODE	nan	<code>BLIND_VALUE_MODE</code>	What counts as a 'blind' LiDAR reading: nan / zero / both.
BLIND_ZERO_EPSILON_M	0.01 m	<code>BLIND_ZERO_EPSILON_M</code>	Threshold below which a reading is treated as zero/blind.
LIDAR_DIAGNOSTIC	False	<code>LIDAR_DIAGNOSTIC</code>	Enable one-off LiDAR diagnostic report at startup.